University of Cantabria

Faculty of Sciences

Final Project

Machine Learning and Data Mining

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1 Introduction

The purpose of this project is to apply the knowledge acquired during the Machine Learning and Data Mining course (2025/2026) to a real-world dataset from the UC Irvine Machine Learning Repository.

The chosen dataset is *Estimation of Obesity Levels Based on Eating Habits and Physical Condition* [3], containing data on individuals from Colombia, Peru and Mexico. It comprises 2,111 records and 16 attributes covering demographics, dietary habits, and physical activity. The target variable `NObeyesdad` encodes seven ordered obesity categories from *Insufficient Weight* to *Obesity Type III.*

The project investigates whether it is possible to predict an individual's weight category from lifestyle and dietary features alone, without relying on direct physical measurements such as weight or height.

All files and figures are available on OneDrive and GitHub.

2 Descriptive Statistics

The dataset has 2,111 records, no missing values. The relevant variables include:

- Demographic and body variables: 'Gender', 'Age', 'Height', 'Weight'
- Eating habits: 'FAVC', 'FCVC', 'NCP', 'CAEC', 'CH2O', 'CALC'
- Lifestyle variables: 'SMOKE', 'SCC', 'FAF', 'TUE', 'MTRANS'
- Obesity category: 'NObeyesdad'

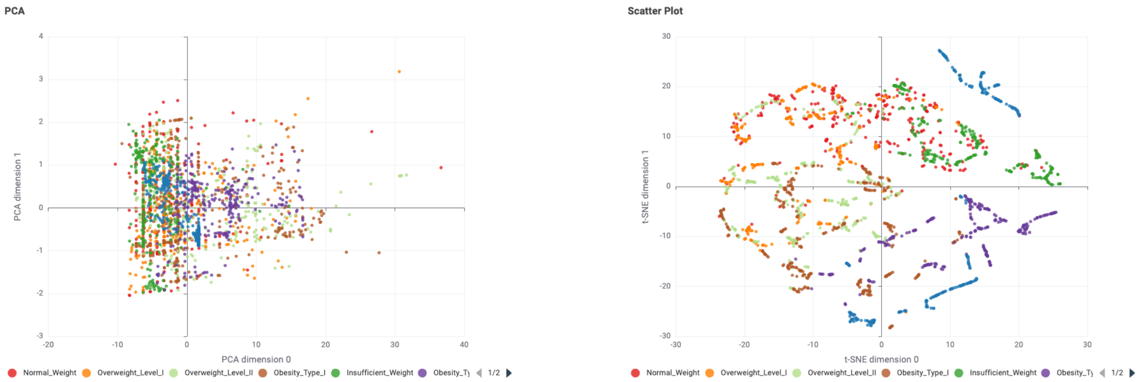
Table 1: Descriptive statistics of continuous variables.

Variable	**Min**	**Max**	**Mean**	**Std. dev.**	**Skewness**	**Kurtosis**
Age	14	61	24.31	6.35	1.53	2.83
Height	1.45	1.98	1.70	0.09	-0.01	-0.56
Weight	39	173	86.59	26.19	0.26	-0.70
FCVC	1	3	2.42	0.53	-0.43	-0.64
NCP	1	4	2.69	0.78	-1.11	0.39
CH2O	1	3	2.01	0.61	-0.10	-0.88
FAF	0	3	1.01	0.85	0.50	-0.62
TUE	0	2	0.66	0.61	0.62	-0.55

The population is predominantly young (mean age 24.3). Physical activity (`FAF`) is moderate on average (1.01/3), and technology usage (`TUE`) is low (0.66/2). The target variable is balanced (each obesity class 13–17%) due to SMOTE oversampling

(77% synthetic, 23% real). Categorical variables show strong skews: `SMOKE=no` in 97.9%, `FAVC=yes` in 88%, `SCC=no` in 95.5%.

Dimensionality reduction visualisations. To better understand the separability of the seven obesity classes, we applied Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) on the normalised feature set (excluding the target). Figure 1 shows both projections side by side. PCA reveals moderate overlap, while t-SNE exhibits more structured clustering, indicating that non-linear methods can capture finer distinctions.



(a) PCA (first two components). (b) t-SNE (perplexity 30).

Figure 1: Dimensionality reduction projections coloured by obesity class.

3 Pre-processing

Encoding. Eight categorical columns were converted to numeric using KNIME's `Category to Number`. `NObeyesdad` was kept as nominal for classifiers.

Exclusion of Height/Weight. `Height` and `Weight` were removed from classification and association rules to avoid data leakage (BMI is directly derived from them).

Normalisation. Continuous variables (`Age`, `FCVC`, `NCP`, `CH2O`, `FAF`, `TUE`) were Z-score normalised.

Train/test split. Stratified 80/20 split preserving class distribution.

Correlation analysis. We examined the covariance matrix (Figure 2) to detect strong linear relationships among features. Most absolute covariances are low (near zero), indicating that the continuous variables are largely independent. This is beneficial for models that assume feature independence and also reduces multicollinearity concerns for regression. Some moderate covariance exists between `FCVC` and `NCP`, as well as between `FAF` and `TUE` (inverse relationship, as expected: more physical activity correlates with less screen time). No extremely high covariances were found, so all features were retained.

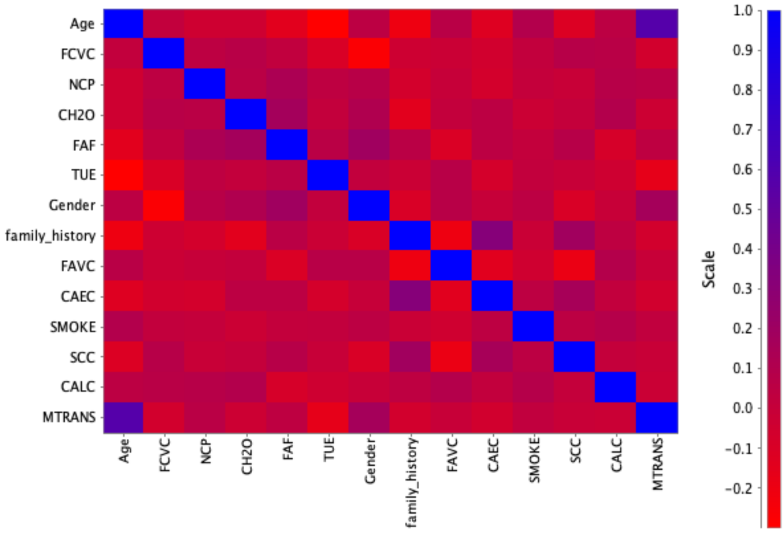


Figure 2: Covariance matrix of the continuous features. Most off-diagonal values are near zero, indicating low linear correlation.

4 Models Used

- **Clustering**: k-means (with elbow method and silhouette coefficient for k selection).
- **Classification**: Logistic Regression, SVM, Decision Tree Random Forest.
- **Neural Networks**: RProp MLP Learner (KNIME native) and H2O AutoML Deep Learning.
- **Association rules**: Apriori algorithm (support 0.05, confidence 0.70).

5 Data Processing

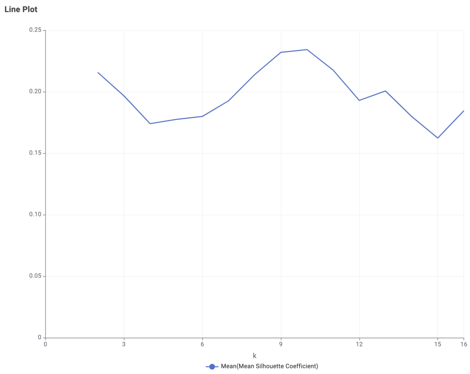
5.1 Clustering

5.1.1 Objective

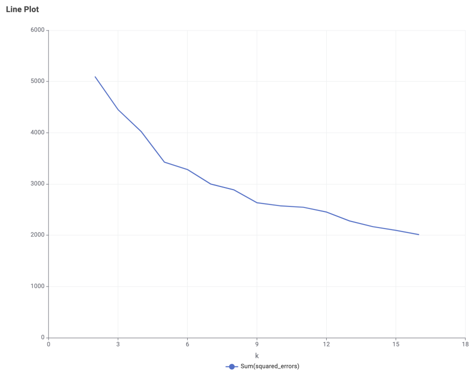
Identify natural groups based on demographics, eating habits, and lifestyle, without using `NObeyesdad` (unsupervised). Afterwards, interpret clusters using the true obesity labels.

5.1.2 Method

K-means with one-hot encoding and normalisation. Tested $k = 2$ to 16. The silhouette coefficient and elbow method were used to select k. Figure 3 presents both validation plots side by side.



(a) Silhouette coefficient.



(b) Elbow method (sum of squared errors).

Figure 3: Cluster validation for k-means. The silhouette peaks at $k = 9, 10$; the elbow flattens after $k = 9$.

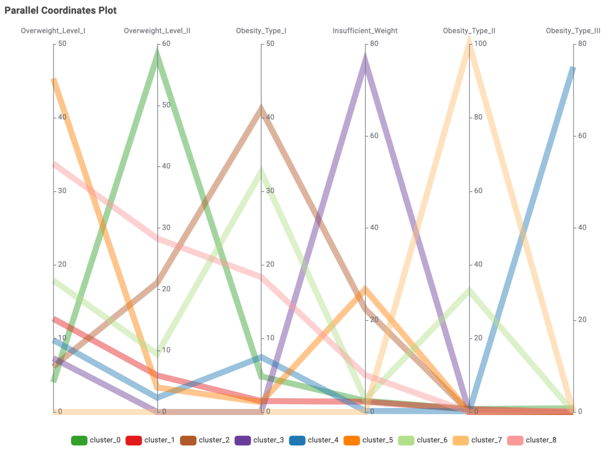
The silhouette coefficient peaked at $k = 9, 10$; the elbow showed diminishing returns after $k = 9$. Therefore $k = 9$ was chosen.

5.1.3 Group Personas

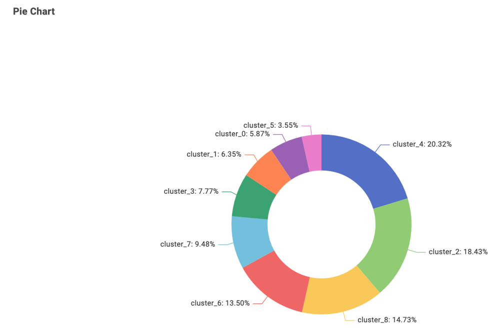
- **Cluster 0 (124, 5.9%)**: Female, Overweight Level II – moderately active.
- **Cluster 1 (134, 6.4%)**: Male, Normal Weight – active, regular meals.
- **Cluster 2 (389, 18.4%)**: Female, Obesity Type I – transitional.
- **Cluster 3 (164, 7.8%)**: Female, Insufficient Weight – young, low hydration.
- **Cluster 4 (429, 20.3%)**: Female, Obesity Type III (75.1%) – sedentary.
- **Cluster 5 (75, 3.6%)**: Female, Overweight Level I – calorie monitoring.
- **Cluster 6 (285, 13.5%)**: Male, Obesity Type II – older, automobile.
- **Cluster 7 (200, 9.5%)**: Male, Obesity Type II (100%) – tall, heavy.
- **Cluster 8 (311, 14.7%)**: Male, Overweight I/II – tall, young.

Even without the label, clusters align strongly with true obesity categories (e.g., Cluster 7 is 100% Obesity Type II), confirming that lifestyle variables capture meaningful structure.

Some clusters are strongly aligned with a single obesity group. Cluster 7 is the clearest case, with 100.0 % of its records classified as Obesity Type II. Cluster 4 is also very strong, with 75.1% Obesity Type III‘, while Cluster 3 contains 76.2% ‘Insufficient Weight‘ and Cluster 1 contains 76.9% ‘Normal Weight‘. These clusters show that k-means successfully discovered several meaningful body-status groups.



(a) Elbow method based on the sum of squared errors.



(b) Distribution of clusters ($k = 9$).

Figure 4: Cluster validation and composition.

5.2 Classification (Classical Methods)

Table 2 shows test set accuracy (80/20 split, height/weight excluded). Random Forest performs best (85.6%).

Table 2: Accuracy of classical classifiers.

Model	Accuracy
Random Forest	85.6%
Decision Tree	76.0%
Logistic Regression	62.2%
SVM	28.6%

The results, summarized in Table 2, show that ensemble methods significantly outperform linear models. Random Forest achieved the highest accuracy (85.6%), indicating that the combination of many decision trees captures the complex, non-linear relationships between lifestyle habits and obesity levels. The Decision Tree alone reached 76.0% accuracy, while Logistic Regression obtained 62.2%, highlighting the non-linear nature of the problem. SVM performed poorly (28.6%), likely due to the high dimensionality and the absence of proper kernel tuning. These results confirm that lifestyle and dietary features contain strong predictive signals, even when height and weight are excluded.

5.3 Neural Networks

5.3.1 RProp MLP Learner

A multilayer perceptron with 3 hidden layers (20 neurons each) and RProp optimization was trained on the same split. The network receives 14 input features (all except `Height`, `Weight`, and the target). Table 3 reports per-class metrics from the KNIME Scorer (423 test samples). Overall accuracy: **72.3%**.

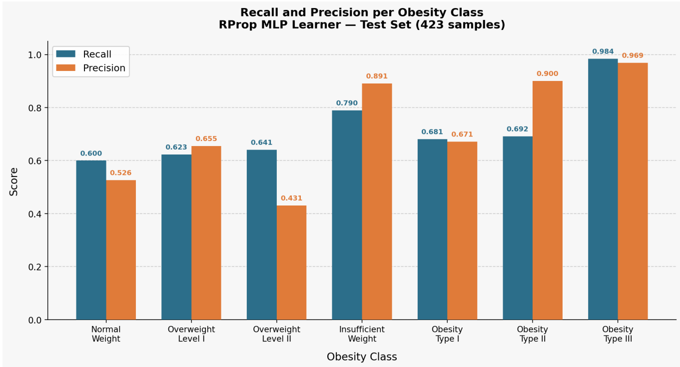
Table 3: Per-class performance of RProp MLP.

Class	Recall	Precision	F1-score	Specificity
Insufficient_Weight	0.790	0.891	0.837	0.971
Normal_Weight	0.600	0.526	0.561	0.920
Overweight_Level_I	0.623	0.655	0.639	0.943
Overweight_Level_II	0.641	0.431	0.515	0.905
Obesity_Type_I	0.681	0.671	0.676	0.945
Obesity_Type_II	0.692	0.900	0.783	0.968
Obesity_Type_III	0.984	0.969	0.976	0.996

Figure 5 presents the confusion matrix and the recall/precision chart side by side. Extreme classes are classified very well; intermediate overweight classes show more confusion, as expected from the ordinal nature of the target.

Table 4: Confusion matrix for the RProp MLP classifier (test set, 423 samples). Rows = actual classes, columns = predicted classes.

	Insufficient Weight	Normal Weight	Overweight I	Overweight II	Obesity I	Obesity II	Obesity III
Insufficient Weight	49						
Normal Weight		30					
Overweight Level I			38				
Overweight Level II				25			
Obesity Type I					47		
Obesity Type II						54	
Obesity Type III							63



(a) Recall and precision per class.

Figure 5: RProp MLP performance visualisations.

5.3.2 H2O AutoML Deep Learning

As a more advanced neural network approach, we used the H2O AutoML framework to automatically search over deep learning architectures (number of layers, neurons, dropout, regularisation). The best model found achieved the metrics shown in Table 5. Overall accuracy: **75.7%**, with Cohen's $\kappa = 0.715$.

Table 5: Per-class performance of H2O AutoML Deep Learning model.

Class	Recall	Precision	F1-score	Specificity
Insufficient_Weight	0.975	0.988	0.981	0.997
Normal_Weight	0.557	0.733	0.633	0.956
Overweight_Level_I	0.935	0.630	0.753	0.959
Overweight_Level_II	0.781	0.893	0.833	0.984
Obesity_Type_I	0.649	0.781	0.709	0.962
Obesity_Type_II	0.733	0.550	0.629	0.932
Obesity_Type_III	0.766	0.653	0.705	0.931

The H2O model outperforms the RProp MLP (75.7% vs. 72.3% accuracy), demonstrating the benefit of automated architecture search and regularisation.

5.4 k-Nearest Neighbors

kNN with Euclidean distance, one-hot encoding of categoricals, and min-max normalisation. The optimal $k = 1$ (Fig. 6) achieved **79.9%** accuracy on the test set (Table 6b). Accuracy decreases monotonically as k increases, suggesting that the nearest neighbour is already highly reliable, likely due to the densification from SMOTE.

Table 6: kNN performance metrics.

(a) Test-set accuracy for different k.

k	Accuracy
1	79.91%
3	78.72%
5	76.83%
7	74.94%
10	73.99%

(b) kNN performance ($k = 1$).

Metric	Value
Accuracy	79.91%
Correct predictions	337 / 422

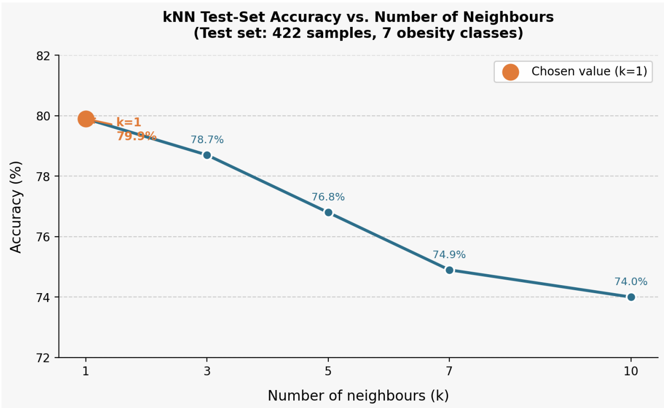


Figure 6: Accuracy vs. k for kNN. The optimum is $k = 1$.

5.5 Association Rules

Apriori algorithm (support 0.05, confidence 0.70) after discretisation of numerical variables (Table 7) and exclusion of height/weight. The six strongest rules are shown in Table 8, followed by a detailed commentary.

Table 7: Discretisation intervals for numerical variables.

Variable	**Bin 1**	**Bin 2**	**Bin 3**
Age	young (≤ 25)	adult (25–45]	senior (> 45)
FAF	sedentary (≤ 0.5)	moderate (0.5–2.5]	active (> 2.5)
TUE	low (≤ 0.6)	medium (0.6–1.3]	high (> 1.3)
FCVC	low (≤ 1.5)	medium (1.5–2.5]	high (> 2.5)
NCP	1–2 (< 2.5)	three (2.5–3.5)	¿3 (≥ 3.5)
CH2O	low (≤ 1.5)	medium (1.5–2.5]	high (> 2.5)

Table 8: Top association rules (sorted by confidence).

# Lift	Antecedent	Consequent	τ	γ
1 5.38	Age=adult, Gender=Female, MTRANS=Public_Transportation	Obesity_Type_III	0.079	0.826
2 5.33	Age=adult, FAF=sedentary, MTRANS=Public_Transportation	Obesity_Type_III	0.079	0.818
3 5.04	Age=adult, FAF=sedentary, FCVC=high	Obesity_Type_III	0.079	0.773
4 5.00	CALC=Sometimes, CH2O=high, Gender=Female	Obesity_Type_III	0.070	0.767
5 4.69	FAF=sedentary, FCVC=high, NCP=three	Obesity_Type_III	0.089	0.719
6 5.15	Age=adult, Gender=Male, MTRANS=Public_Transportation	Obesity_Type_II	0.055	0.725

Commentary:

Rule 1 (highest confidence): Adult women using public transport have an 82.6% probability of Obesity Type III (lift 5.38). This is the strongest association.

Rule 2: Replacing gender with sedentariness (FAF=sedentary) gives nearly identical confidence (81.8%), highlighting the combined effect of physical inactivity and passive transport.

Rule 3: High vegetable consumption (FCVC=high) combined with sedentariness still leads to severe obesity – a counterintuitive result that likely reflects correlation, not causation, and may be amplified by synthetic SMOTE data.

Rule 4: Women who occasionally drink alcohol and drink plenty of water show a high probability of Obesity Type III. This again underlines that association rules capture correlations, not necessarily healthy behaviours.

Rule 5: The rule with highest support (8.9%) confirms that sedentariness dominates even when meal frequency is normal (three meals) and vegetable intake is high.

Rule 6: The only rule for Obesity Type II. Adult men using public transport have a 72.5% probability of Type II (compared to Type III for women), suggesting gender-specific patterns.

No rules were found for normal weight or overweight classes – those categories have less pronounced behavioural signatures.

6 Results Obtained

- **Clustering**: k-means with $k = 9$ produced groups that strongly align with true obesity labels. Cluster 7 is 100% Obesity Type II; Cluster 4 is 75% Obesity Type III; Cluster 3 is 76% Insufficient Weight; Cluster 1 is 77% Normal Weight.
- **Classical classification**: Random Forest achieved 85.6% accuracy, Decision Tree 76.0%, Logistic Regression 62.2%, SVM 28.6%.
- **Neural networks**: RProp MLP reached 72.3% accuracy; H2O AutoML improved to 75.7%, showing the value of automated architecture search.
- **k-Nearest Neighbors**: 79.9% accuracy with $k = 1$, indicating well-separated local neighbourhoods in the feature space.
- **Association rules**: The most predictive pattern for severe obesity is *adult female, public transport user, sedentary* (confidence 82.6%, lift 5.38). Sedentariness appears repeatedly as a dominant factor.
- **Limitations**: 77% of the data is synthetic (SMOTE), which may overestimate performance. Computational time prevented lowering the support threshold for association rules.

7 Conclusion

This project successfully applied clustering, classification, and association rule mining to the obesity dataset. Key conclusions:

- Lifestyle and dietary habits alone are sufficient to predict obesity levels with good accuracy (up to 85.6% with Random Forest, 79.9% with kNN). This validates that non-clinical, self-reported data can be used for risk assessment.
- Clustering without the target label still recovers the true obesity categories, confirming that the chosen features contain strong structural information.
- Sedentariness is the most influential factor associated with severe obesity, outweighing some healthy eating habits (e.g., high vegetable consumption).
- The synthetic nature of most data (SMOTE) is a limitation; future work should collect more real data and explore deeper architectures (e.g., transformers) to improve generalisation and causal inference.

Overall, the project demonstrates that data mining provides valuable tools for understanding and predicting obesity from everyday lifestyle information.

References

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